

ECON 6000 – Microeconomics I

Assignment 1

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Question 1

We have $L = 2$, $x(p, w)$ is single-valued, and $x(p, w)$ satisfies Walras' law and WARP.

1.1

Yes, the Slutsky matrix, S , is symmetric when $L = 2$. Since there are only 2 commodities, we end up with a 2×2 Slutsky matrix, so to prove symmetry of S we essentially need to prove $S_{12} = S_{21}$.

Firstly, since $x(p, w)$ is single-valued and we assume it is differentiable, all the derivatives below exist, so the Slutsky matrix is well defined. Also, $x(p, w)$ is homogeneous of degree 0 by Walras' law and the Weak Axiom of Revealed Preference. If the function is homogeneous of degree 0, we can later apply the results from Prop. 2.E.1. Let's write down the (l, k) th entry of S :

$$S_{lk}(p, w) = \frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w).$$

Step 1 (rows). Multiplying by p_k ,

$$S_{lk} p_k = \frac{\partial x_l(p, w)}{\partial p_k} p_k + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w) p_k.$$

Summing across k and applying the result from Prop. 2.E.1 (homogeneity of degree 0, which we get from WARP and Walras' law), we obtain

$$\begin{aligned} \sum_k S_{lk} p_k &= \sum_k \frac{\partial x_l(p, w)}{\partial p_k} p_k + \frac{\partial x_l(p, w)}{\partial w} \sum_k x_k(p, w) p_k \\ &= \sum_k \frac{\partial x_l(p, w)}{\partial p_k} p_k + \frac{\partial x_l(p, w)}{\partial w} w = 0, \end{aligned}$$

where the second line uses Walras' law, $\sum_k p_k x_k(p, w) = w$. Hence

$$S_{l1} p_1 + S_{l2} p_2 = 0 \quad \forall l = 1, 2.$$

In particular, for $l = 1$:

$$S_{11} p_1 + S_{12} p_2 = 0 \implies S_{12} = \frac{-S_{11} p_1}{p_2}.$$

Step 2 (columns). Next, multiplying by p_l ,

$$p_l S_{lk}(p, w) = \frac{\partial x_l(p, w)}{\partial p_k} p_l + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w) p_l.$$

Summing over all goods l and applying Props. 2.E.2 and 2.E.3 (both follow from differentiating Walras' law),

$$\begin{aligned} \sum_l S_{lk}(p, w) p_l &= \sum_l \frac{\partial x_l(p, w)}{\partial p_k} p_l + x_k(p, w) \sum_l \frac{\partial x_l(p, w)}{\partial w} p_l \\ &= \underbrace{-x_k(p, w)}_{(1)} + \underbrace{x_k(p, w)}_{(2)} = 0 \quad \forall k = 1, 2. \end{aligned}$$

- (1) comes from Prop. 2.E.2 (Cournot aggregation): $\sum_l p_l \frac{\partial x_l(p, w)}{\partial p_k} + x_k(p, w) = 0$, i.e.

$$\sum_l p_l \frac{\partial x_l(p, w)}{\partial p_k} = -x_k(p, w).$$

- (2) comes from Prop. 2.E.3 (Engel aggregation): $\sum_l p_l \frac{\partial x_l(p, w)}{\partial w} = 1$.

Therefore $\sum_l S_{lk} p_l = 0$ for all $k = 1, 2$. In particular, for $k = 1$:

$$S_{11}p_1 + S_{21}p_2 = 0 \implies S_{21} = \frac{-S_{11}p_1}{p_2} = S_{12}.$$

Therefore the Slutsky matrix is symmetric for $L = 2$.

1.2

No, the proof does not extend to the case where $L = 3$. Recall from 1.1 that

$$S(p, w)p = 0 \quad (1) \quad \text{and} \quad S(p, w)^T p = 0 \quad (2).$$

Taking the first row of each,

$$S_{11}p_1 + S_{12}p_2 + S_{13}p_3 = 0 \quad \text{from (1)} \quad (3)$$

$$S_{11}p_1 + S_{21}p_2 + S_{31}p_3 = 0 \quad \text{from (2)} \quad (4)$$

Subtracting, (3) – (4) gives

$$(S_{12} - S_{21})p_2 + (S_{13} - S_{31})p_3 = 0.$$

Since $p_2, p_3 > 0$, we may have the case that

$$S_{12} - S_{21} = p_3 \quad \text{and} \quad S_{13} - S_{31} = -p_2,$$

so that $p_3p_2 - p_2p_3 = 0$. In this case $S_{12} \neq S_{21}$ and $S_{13} \neq S_{31}$. In $L = 2$ there is only one term left, so the equation forces $S_{12} = S_{21}$; with two terms it no longer does. Therefore, the conditions from 1.1 no longer force symmetry, and the proof does not extend to $L = 3$.

1.3

SARP is just a stronger version of WARP that forbids preference reversal (also through indirect chains). According to Prop. 2.F.2, if $x(p, w)$ satisfies Walras' law and WARP, then $S(p, w)$ is negative semidefinite. Since SARP implies WARP, this holds under SARP as well. For $L = 2$, we know from 1.1 that S is also symmetric, so we agree with the statement. However, the argument does not stand for $L = 3$. SARP still gives negative semidefiniteness through WARP, but by 1.2 the tools from Chapter 2 cannot deliver symmetry when $L = 3$. So with only what we know from Chapter 2, we cannot conclude that the statement holds for $L = 3$. This does not mean S fails to be symmetric; we just cannot prove it here.

Question 2

2.1

Let $w^t = p^t \cdot x^t$ denote the wealth in year t , so $w^1 = 18$, $w^2 = 10$, $w^3 = 15$.

x^1 vs. x^2 :

$$p^1 \cdot x^2 = 15 < p^1 \cdot x^1 = 18, \quad p^2 \cdot x^1 = 15 > p^2 \cdot x^2 = 10.$$

So x^1 is revealed preferred to x^2 , and not the other way around.

x^3 vs. x^2 :

$$p^3 \cdot x^2 = 10 < p^3 \cdot x^3 = 15, \quad p^2 \cdot x^3 = 15 > p^2 \cdot x^2 = 10.$$

So x^3 is revealed preferred to x^2 .

x^3 vs. x^1 :

$$p^3 \cdot x^1 = 15 = p^3 \cdot x^3, \quad p^1 \cdot x^3 = 27 > p^1 \cdot x^1 = 18.$$

So x^3 is revealed preferred to x^1 .

Hence x^3 is revealed preferred to x^1 , x^1 is revealed preferred to x^2 , and x^3 is revealed preferred to x^2 . In every pair, the bundle that is revealed preferred is not affordable when the other bundle is chosen, so there is no preference reversal and WARP is satisfied.

2.2

From the previous results, we already know $x^3 \succsim x^2$. Now we need to compare x^4 with x^3 , and x^4 with x^2 .

x^4 and x^3 :

$$p^3 \cdot x^3 = 15 > p^3 \cdot x^4 = 14,$$

then $x^3 \succ x^4$. By monotonicity, we may find $x^5 = x^4 + (0.2, 0.5) = (4.2, 10.5) \gg x^4$, therefore $x^5 \succ x^4$. But

$$p^3 \cdot x^5 = 4.2 \times 1 + 10.5 \times 1 = 14.7 < w^3 = 15,$$

so $x^3 \succ x^5 \succ x^4 \implies x^3 \succ x^4$.

x^4 and x^2 :

$$p^2 \cdot x^4 = 14 > w^2 = 10,$$

so it's inconclusive from WARP. Since $x^3 \succsim x^2$, by convexity

$$\bar{x} = \alpha x^3 + (1 - \alpha)x^2 \succsim x^2, \quad \alpha \in [0, 1].$$

Choose α so that $\bar{x}$ lies strictly below x^4 in both goods; $\alpha = 0.6$ works:

$$\bar{x} = 0.4x^2 + 0.6x^3 = (3.8, 9.2) \succsim x^2.$$

Since $x^4 = (4, 10) \gg \bar{x}$, by monotonicity $x^4 \succ \bar{x}$, and then $x^4 \succ x^2$. By transitivity,

$$x^3 \succ x^4 \succ x^2.$$

So x^4 is worse than x^3 but better than x^2 .

With convexity only. We may still conclude $x^3 \succsim x^4$ since x^3 is revealed preferred to x^4 (as $p^3 \cdot x^4 < w^3$), and we may still say $\alpha x^3 + (1 - \alpha)x^4 \succsim x^4$, but no further conclusion can be made without the monotonicity assumption. For x^4 and x^2 , we may still say $\bar{x} = (3.8, 9.2) \succsim x^2$. Even though $x^4 \gg \bar{x}$, without monotonicity we cannot conclude $x^4 \succ \bar{x} \succsim x^2$, so x^4 and x^2 cannot be compared. The comparison collapses.

2.3

Now $x^4 = (4, 10)$ is chosen at $p^4 = (2, 1)$, so $w^4 = p^4 \cdot x^4 = 18$. The expanded table is:

	$p^1 = (1, 2)$	$p^2 = (1, 1)$	$p^3 = (1, 1)$	$p^4 = (2, 1)$
$x^1 = (12, 3)$	18	15	15	27
$x^2 = (5, 5)$	15	10	10	15
$x^3 = (3, 12)$	27	15	15	18
$x^4 = (4, 10)$	24	14	14	18

We can still obtain the previous results: x^1 is revealed preferred to x^2 , x^3 is revealed preferred to x^2 , and x^3 is revealed preferred to x^1 .

Check x^4 with x^3 :

$$p^3 \cdot x^4 = 14 < w^3 = 15 \quad \text{but} \quad p^4 \cdot x^3 = 18 = w^4 = p^4 \cdot x^4.$$

So x^3 is revealed preferred to x^4 , and x^4 is also revealed preferred to x^3 . WARP is violated.

Check x^4 with x^2 :

$$p^4 \cdot x^2 = 15 < p^4 \cdot x^4 = 18 \quad \text{and} \quad p^2 \cdot x^4 = 14 > p^2 \cdot x^2 = 10,$$

so it has to be that x^4 is revealed preferred to x^2 .

Check x^4 with x^1 : $p^4 \cdot x^1 = 27 > w^4$ and $p^1 \cdot x^4 = 24 > w^1$, so neither is revealed preferred to the other.

In summary, x^1 is revealed preferred to x^2 , x^3 to x^2 , x^3 to x^1 , and x^4 to x^2 . But x^3 is revealed preferred to x^4 and x^4 is also revealed preferred to x^3 , so WARP is not satisfied.

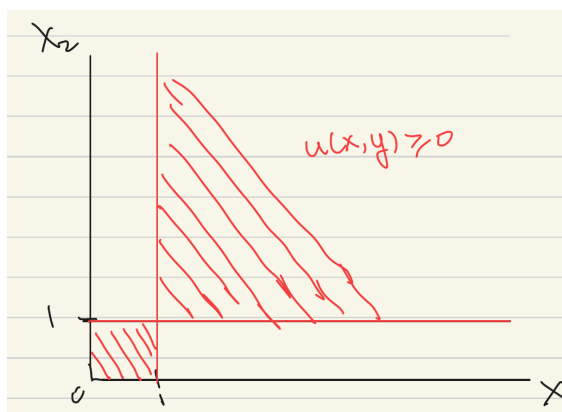
Question 3

We assume the standard non-negative consumption set $X = \mathbb{R}_+^2$, and

$$u(x_1, x_2) = (x_1 - 1)(x_2 - 1).$$

3.1

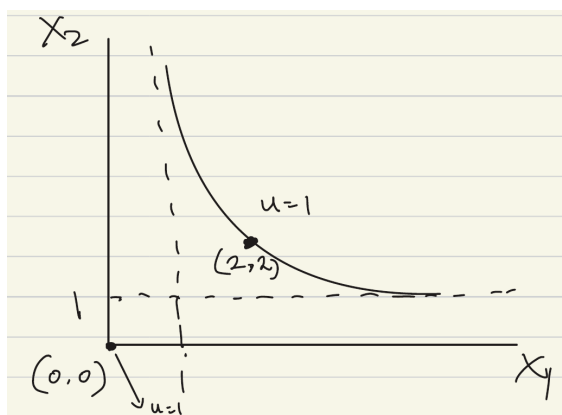
Utility level 0. $u(x_1, x_2) = 0$ exactly when $x_1 = 1$ or $x_2 = 1$, so the indifference set is the two lines $x_1 = 1$ and $x_2 = 1$. The superior set $\{x : u(x) \geq 0\}$ consists of two pieces: the region where $x_1 \geq 1$ and $x_2 \geq 1$ (both factors non-negative), and the box $[0, 1]^2$, where both factors are non-positive so their product is still non-negative. The two remaining regions ($x_1 > 1, x_2 < 1$ and $x_1 < 1, x_2 > 1$) give $u < 0$. The hatched area below is the superior set.



Utility level 1. When $u = 1$,

$$(x_1 - 1)(x_2 - 1) = 1 \implies x_2 - 1 = \frac{1}{x_1 - 1} \implies x_2 = \frac{1}{x_1 - 1} + 1 = \frac{x_1}{x_1 - 1}.$$

As $x_1 \rightarrow \infty$, $x_2 \rightarrow 1$, and as $x_1 \rightarrow 1^+$, $x_2 \rightarrow \infty$, so the curve has asymptotes $x_1 = 1$ and $x_2 = 1$ and passes through $(2, 2)$. In addition, $u(0, 0) = (-1)(-1) = 1$, so the origin is also on this indifference set. It is the only such point in $[0, 1]^2$, since there $(1 - x_1)(1 - x_2) < 1$ unless $x_1 = x_2 = 0$.



3.2

Not convex. For a preference relation to be convex, every upper contour set $\{(x_1, x_2) : (x_1 - 1)(x_2 - 1) \geq k\}$ must be convex. Take $k = 1$ and consider the two points $(0, 0)$ and $(2, 2)$, both with $u = 1$. Their convex combination

$$0.5(0, 0) + 0.5(2, 2) = (1, 1), \quad u(1, 1) = 0 < 1 = u(0, 0) = u(2, 2),$$

has a lower utility level, so $(1, 1)$ is not in the upper contour set. Hence the upper contour set is not convex and the preferences are not convex.

Not locally non-satiated. Take $x = (0, 0)$ and any $0 < \varepsilon < 1$. For every $y \in B_\varepsilon(0, 0) \cap \mathbb{R}_+^2$ with $y \neq (0, 0)$ we have $0 \leq y_1, y_2 < 1$ with at least one strictly positive, so

$$u(y) = (1 - y_1)(1 - y_2) < 1 = u(0, 0).$$

There is no bundle arbitrarily close to $(0, 0)$ that is strictly preferred to it, so local non-satiation fails at $(0, 0)$.

Not monotone. Monotonicity implies local non-satiation, so the preferences cannot be monotone if they are not locally non-satiated. Directly: $(0.5, 0.5) \gg (0, 0)$ but $u(0.5, 0.5) = 0.25 < 1 = u(0, 0)$.

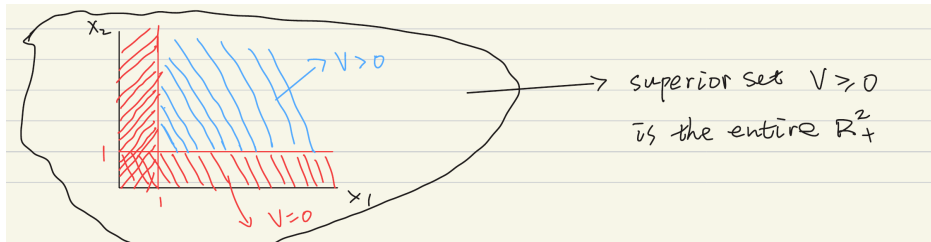
3.3

Now

$$v(x_1, x_2) = \begin{cases} (x_1 - 1)(x_2 - 1) & \text{if } x_1 > 1 \text{ and } x_2 > 1, \\ 0 & \text{otherwise (i.e. } x_1 \leq 1 \text{ or } x_2 \leq 1), \end{cases}$$

with $x_1, x_2 \geq 0$.

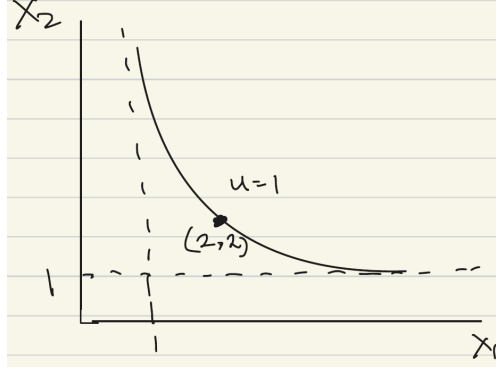
Utility level 0. The indifference set $v = 0$ is the whole region where $x_1 \leq 1$ or $x_2 \leq 1$ (red below), not just two lines. On the region $x \gg 1$ (blue) we have $v > 0$. Therefore the superior set $\{x : v(x) \geq 0\}$ is the entire $\mathbb{R}_+^2$, as indicated by the arrow in the figure.



Utility level 1. When $v = 1$ it must be the case that $x \gg 1$, and as before

$$x_2 = \frac{x_1}{x_1 - 1}.$$

The graph is the same as in 3.1 except that $(0, 0)$ is no longer included, since $v(0, 0) = 0$. (The curve in the figure is $v = 1$.)



3.4

Convex. The preference represented by v is convex. For $k \leq 0$, the upper contour set $\{x : v(x) \geq k\}$ is the entire $\mathbb{R}_+^2$, which is convex. For $k > 0$, the upper contour set is

$$\{x \in \mathbb{R}_+^2 : x \gg 1, (x_1 - 1)(x_2 - 1) \geq k\} = \left\{x : x_1 > 1, x_2 \geq 1 + \frac{k}{x_1 - 1}\right\},$$

the region on and above the upper hyperbola branch. The boundary $g(x_1) = 1 + \frac{k}{x_1 - 1}$ satisfies $g''(x_1) = \frac{2k}{(x_1 - 1)^3} > 0$ for $x_1 > 1$, so it is a convex function, and the region above the graph of a convex function on the convex domain $x_1 > 1$ is a convex set. All upper contour sets are convex, so the preferences are convex.

Not locally non-satiated. For $x = (0, 0)$ and any $0 < \varepsilon < 1$, every $y \in B_\varepsilon(0, 0) \cap \mathbb{R}_+^2$ has $y_1, y_2 < 1$, so $v(y) = 0 = v(0, 0)$. No nearby bundle is strictly preferred, so local non-satiation fails.

Not monotone. As before, this implies the preferences are not monotone. Directly: $(0.5, 0.5) \gg (0, 0)$ but $v(0.5, 0.5) = 0 = v(0, 0)$.

3.5

We assume $p_1 + p_2 < w$, so $(1, 1)$ is affordable, and ask whether the same bundle x^* is optimal for both $v(x_1, x_2)$ and $u(x_1, x_2)$.

Starting with v . At $(1, 1)$ we only obtain $v(1, 1) = 0$. Since $p_1 + p_2 < w$, we may still increase both goods by a small amount so that $x_1, x_2 > 1$, which gives a strictly positive utility while remaining affordable. So the optimum lies in the region $x \gg 1$. There v is strictly increasing in both arguments, so the budget constraint binds, and we solve

$$\max_x v(x) = (x_1 - 1)(x_2 - 1) \quad \text{s.t. } x \gg 1 \text{ and } p_1 x_1 + p_2 x_2 = w.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = (x_1 - 1)(x_2 - 1) + \lambda(w - p_1 x_1 - p_2 x_2),$$

with first-order conditions

$$\mathcal{L}_{x_1} = x_2 - 1 - \lambda p_1 = 0, \quad \mathcal{L}_{x_2} = x_1 - 1 - \lambda p_2 = 0 \implies \frac{x_2 - 1}{x_1 - 1} = \frac{p_1}{p_2}.$$

So $(x_2 - 1)p_2 = (x_1 - 1)p_1$, i.e. $x_1 = \frac{(x_2 - 1)p_2}{p_1} + 1$. From the budget constraint, $x_1 = \frac{w - p_2 x_2}{p_1}$. Setting these equal,

$$w - p_2 x_2 = (x_2 - 1)p_2 + p_1 \implies 2p_2 x_2 = w + p_2 - p_1 \implies x_2^* = \frac{w + p_2 - p_1}{2p_2},$$

and similarly

$$x_1^* = \frac{w + p_1 - p_2}{2p_1}.$$

Both exceed 1 because $w > p_1 + p_2$. The maximized utility is

$$\begin{aligned} v(x^*) &= (x_1^* - 1)(x_2^* - 1) = \left(\frac{w + p_1 - p_2}{2p_1} - 1 \right) \left(\frac{w + p_2 - p_1}{2p_2} - 1 \right) \\ &= \left(\frac{w - p_1 - p_2}{2p_1} \right) \left(\frac{w - p_1 - p_2}{2p_2} \right) = \frac{(w - p_1 - p_2)^2}{4p_1 p_2} > 0. \end{aligned}$$

For u , it is different. There are two candidates:

- On $[0, 1]^2$, $u(x) = (1 - x_1)(1 - x_2)$ reaches its maximum at $(0, 0)$, with $u = 1$, and this bundle costs nothing.
- On $x \gg 1$, u coincides with v , so the best choice there is the same x^* , with $u(x^*) = \frac{(w - p_1 - p_2)^2}{4p_1 p_2}$.

In the remaining regions (exactly one good below 1) we have $u \leq 0$, so they are never optimal.

When they do not share the same bundle. If

$$\begin{aligned} \frac{(w - p_1 - p_2)^2}{4p_1 p_2} < 1 &\iff (w - p_1 - p_2)^2 < 4p_1 p_2 \iff w - p_1 - p_2 < 2\sqrt{p_1 p_2} \\ &\iff p_1 + p_2 < w < p_1 + p_2 + 2\sqrt{p_1 p_2}, \end{aligned}$$

then the consumer with u chooses $(0, 0)$, while the consumer with v chooses x^* . Otherwise, if $w > p_1 + p_2 + 2\sqrt{p_1 p_2}$, the u consumer also chooses $x \gg 1$, so both solve the same optimization problem and have the same optimal bundle x^* (at equality, u is indifferent between $(0, 0)$ and x^*). Therefore we do not agree in general: the same bundle is optimal for both only when wealth is high enough.

Question 4

4.1

Yes, the preferences are strictly convex. Recall that $\succsim$ is strictly convex if for every $x \in X$,

$$y \succsim x, \quad z \succsim x, \quad y \neq z \implies \alpha y + (1 - \alpha)z \succ x \quad \forall \alpha \in (0, 1).$$

Step 1 (u is strictly concave). Take any $y, z \in X = \mathbb{R}_+^n$ with $y \neq z$ and any $\alpha \in (0, 1)$. Since X is convex, $\alpha y + (1 - \alpha)z \in X$. Split the goods into two sets:

$$I = \{i : y_i \neq z_i\}, \quad J = \{j : y_j = z_j\}.$$

Since $y \neq z$, the set I is non-empty.

For $i \in I$, strict concavity of f_i gives

$$f_i(\alpha y_i + (1 - \alpha)z_i) > \alpha f_i(y_i) + (1 - \alpha)f_i(z_i). \quad (\text{a})$$

For $j \in J$, we have $\alpha y_j + (1 - \alpha)z_j = y_j = z_j$, so

$$f_j(\alpha y_j + (1 - \alpha)z_j) = \alpha f_j(y_j) + (1 - \alpha)f_j(z_j). \quad (\text{b})$$

Summing (a) over $i \in I$ and (b) over $j \in J$, and using that I contains at least one good, the sum keeps the strict inequality:

$$u(\alpha y + (1 - \alpha)z) = \sum_{i=1}^n f_i(\alpha y_i + (1 - \alpha)z_i) > \alpha \sum_{i=1}^n f_i(y_i) + (1 - \alpha) \sum_{i=1}^n f_i(z_i) = \alpha u(y) + (1 - \alpha)u(z).$$

Hence u is strictly concave on X .

Step 2 (strict convexity of $\succsim$). Now let $y \succsim x$ and $z \succsim x$ with $y \neq z$, i.e. $u(y) \geq u(x)$ and $u(z) \geq u(x)$. For any $\alpha \in (0, 1)$, Step 1 gives

$$u(\alpha y + (1 - \alpha)z) > \alpha u(y) + (1 - \alpha)u(z) \geq \alpha u(x) + (1 - \alpha)u(x) = u(x).$$

Therefore $\alpha y + (1 - \alpha)z \succ x$, and the preferences represented by (1) are strictly convex.

4.2

Case $f_1(x_1) = x_1$. No, the answer does not change: the preferences are still strictly convex. However, u is no longer strictly concave, so the proof needs one extra step.

Take any $y, z \in \mathbb{R}_+^n$ with $y \neq z$ and any $\alpha \in (0, 1)$. In the case where y and z differ in at least one good other than good 1, the set I from 4.1 contains a good whose f_i is strictly concave, so the sum keeps the strict inequality and the proof from 4.1 stays the same.

However, if we consider y and z that only differ in good 1, i.e. $y_1 \neq z_1$ and $y_j = z_j$ for all $j \geq 2$, then for good 1 we have

$$f_1(\alpha y_1 + (1 - \alpha)z_1) = \alpha y_1 + (1 - \alpha)z_1 = \alpha f_1(y_1) + (1 - \alpha)f_1(z_1),$$

and every other good satisfies (b). Adding them together,

$$u(\alpha y + (1 - \alpha)z) = \sum_{i=1}^n \left[\alpha f_i(y_i) + (1 - \alpha)f_i(z_i) \right] = \alpha u(y) + (1 - \alpha)u(z).$$

Here, u is only concave (it is linear along good 1), so Step 2 from 4.1 does not go through directly. But since y and z only differ in good 1,

$$u(y) - u(z) = y_1 - z_1 \neq 0,$$

so $u(y) \neq u(z)$. Without loss of generality let $u(y) > u(z)$. Now pick any x with $y \succsim x$ and $z \succsim x$, i.e. $u(y), u(z) \geq u(x)$. Then

$$u(\alpha y + (1 - \alpha)z) = \alpha u(y) + (1 - \alpha)u(z) > \alpha u(z) + (1 - \alpha)u(z) = \min\{u(y), u(z)\} \geq u(x).$$

Therefore $\alpha y + (1 - \alpha)z \succ x$ in every case, and the preference is still strictly convex.

Case $f_1(x_1) = x_1$ and $f_2(x_2) = x_2$. Yes, now the answer changes: the preference is no longer strictly convex. It is enough to find one counterexample.

Fix any $(c_3, \dots, c_n) \in \mathbb{R}_+^{n-2}$ and let

$$y = (1, 0, c_3, \dots, c_n), \quad z = (0, 1, c_3, \dots, c_n),$$

so $y \neq z$. Therefore,

$$u(y) = u(z) = 1 + \sum_{i \geq 3} f_i(c_i).$$

Set $x = z$, so $u(x) = u(z)$, and we have $y \succsim x$ and $z \succsim x$. For any $\alpha \in (0, 1)$, $\alpha y + (1 - \alpha)z = (\alpha, 1 - \alpha, c_3, \dots, c_n)$, and we apply the definition:

$$u(\alpha y + (1 - \alpha)z) = \alpha + (1 - \alpha) + \sum_{i \geq 3} f_i(c_i) = 1 + \sum_{i \geq 3} f_i(c_i) = u(x).$$

So $\alpha y + (1 - \alpha)z \sim x$, not $\alpha y + (1 - \alpha)z \succ x$. Here, the trick from the first case does not work, since y and z can trade good 1 for good 2 and keep $u(y) = u(z)$. Therefore, the preference is convex but not strictly convex.